

Day 15: Clean Account Holder Name

1. Day Number

Day 15

2. Topic Name

String Methods

Today you will learn these useful string methods:

- `strip()`
- `title()`
- `lower()`
- `upper()`

3. Connection

Earlier, you created an account holder name using user input.

However, users may enter names in messy formats:

```
rahul tiwari
```

or:

```
RAHUL TIWARI
```

Today, you will clean and format that input before displaying it.

4. Important Topics

strip()

Removes extra spaces from the **beginning and end** of a string.

```
name = "   Rahul Tiwari   "
clean_name = name.strip()
```

Result:

```
Rahul Tiwari
```

It does not remove normal spaces between words.

title()

Changes the first letter of every word to uppercase.

```
name = "rahul tiwari"
formatted_name = name.title()
```

Result:

```
Rahul Tiwari
```

lower()

Changes all letters to lowercase.

```
name = "RAHUL TIWARI"  
print(name.lower())
```

Result:

```
rahul tiwari
```

upper()

Changes all letters to uppercase.

```
name = "Rahul Tiwari"  
print(name.upper())
```

Result:

```
RAHUL TIWARI
```

5. Foundational Notes

Strings are immutable

Python strings cannot be changed directly.

A string method creates and returns a **new string**.

For example:

```
name = "  rahul  "  
name.strip()
```

The original name still contains spaces because the result was not saved.

You should save the returned value:

```
name = name.strip()
```

Methods can be combined

You can apply more than one string method:

```
name = "  rahul tiwari  "  
clean_name = name.strip().title()
```

Python processes this from left to right:

1. `strip()` removes outside spaces.
2. `title()` formats the words.

Result:

```
Rahul Tiwari
```

`strip()` removes only outside spaces

Consider:

```
Rahul   Tiwari
```

After using `strip()`:

```
Rahul   Tiwari
```

The spaces between `Rahul` and `Tiwari` remain unchanged.

6. Easy Example

```
city = "   bengaluru   "

clean_city = city.strip()
formatted_city = clean_city.title()

print(formatted_city)
```

Output:

```
Bengaluru
```

You could also write:

```
formatted_city = city.strip().title()
```

Both approaches work.

7. Problem Statement

Create a Python program that:

1. Asks the user to enter the account holder name.
2. Accepts a name that may contain extra spaces.
3. Cleans the name using string methods.
4. Formats the name properly.
5. Prints the cleaned account holder name.

Example messy input:

```
rAHUL tiWARI
```

Expected formatted name:

```
Rahul Tiwari
```

8. Concepts Used

You will use:

- Variable
- String
- `input()`
- Function
- Parameter
- Return value
- `strip()`
- `title()`
- Function call
- `print()`

9. Thought Process

Think through the problem in this order:

Step 1: Get the name

Use `input()` to ask the user for the account holder name.

The input may look like:

```
rAHUL tiWARI
```

Step 2: Send the name to a function

Create a function that accepts the messy name as a parameter.

Step 3: Remove extra outside spaces

Use `strip()` to remove spaces from the beginning and end.

After cleaning:

```
rAHUL tiWARI
```

Step 4: Format each word

Use `title()` to make the first letter of every word uppercase.

After formatting:

```
Rahul Tiwari
```

Step 5: Return the cleaned name

The function should return the formatted value.

Step 6: Print the result

Call the function and display the returned name.

10. Pseudocode

```
START

DEFINE a function that receives account holder name

    REMOVE spaces from the beginning and end

    FORMAT the first letter of each word as uppercase

    RETURN the cleaned name

ASK user to enter account holder name

CALL the cleaning function

SAVE the returned value

PRINT the formatted account holder name

END
```

11. Suggested Solving Approach: Functional Approach

Use one small function whose only responsibility is cleaning the name.

Suggested structure:

```
clean_account_holder_name(name)
```

The function should:

1. Receive the name.
2. Apply the required string methods.
3. Return the cleaned name.

The main part of the program should:

1. Take user input.
2. Call the function.
3. Print the returned value.

This keeps the input, processing, and output responsibilities clear.

12. Easy Edge Cases

Extra spaces

Input:

Rahul Tiwari

The spaces before the name should be removed.

All lowercase

Input:

rahul tiwari

Formatted result:

Rahul Tiwari

All uppercase

Input:

RAHUL TIWARI

Formatted result:

Rahul Tiwari

Mixed uppercase and lowercase

Input:

rAHuL tIWArI

Formatted result:

Rahul Tiwari

Spaces at both ends

Input:

rahul tiwari

Formatted result:

Rahul Tiwari

13. Expected Input

Example 1:

```
Enter account holder name:   rahul tiwari
```

Example 2:

```
Enter account holder name: RAHUL TIWARI
```

Example 3:

```
Enter account holder name:   rAHuL tIWArI
```

14. Expected Output

For all the previous examples, the output should be similar to:

```
Account Holder Name: Rahul Tiwari
```

15. Hint Only

Create a function that accepts the name as a parameter. Inside the function, combine two string methods:

```
name.strip().title()
```

Remember to **return** the cleaned value and save the returned result when calling the function.